Review: Identifying congruent triangles

ANSWER KEY

Core Concepts: Review: Identifying congruent triangles

1 In $\triangle ABC$ studying review: identifying congruent triangles, $m\angle A = 30^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 105° ✓

b. 115°

c. 97°

d. 150°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (30^\circ + 45^\circ) = 105^\circ$.

2 A geometric figure scaled by factor $k = 2$ has an original side length of 4 cm. What is the corresponding image length?

a. 6 cm

b. 8 cm ✓

c. 12 cm

d. 16 cm

Multiplying the original length by the scale factor gives $4 \times 2 = 8$ cm.

3 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 7)$. What is its y-coordinate when $x = 1$?

a. 7

b. 12

c. 10 ✓

d. 13

Substituting $x = 1$ into $y = 3x + 7$ gives $y = 3(1) + 7 = 10$.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

Procedural Fluency & Algebraic Manipulation

4 In $\triangle ABC$ studying review: identifying congruent triangles, $m\angle A = 30^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 98° ✓

b. 108°

c. 90°

d. 150°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (30^\circ + 52^\circ) = 98^\circ$.

5 A geometric figure scaled by factor $k = 3$ has an original side length of 4 cm. What is the corresponding image length?

a. 9 cm

b. 12 cm ✓

c. 16 cm

d. 36 cm

Multiplying the original length by the scale factor gives $4 \times 3 = 12$ cm.

6 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 7)$. What is its y-coordinate when $x = 2$?

a. 10

b. 15

c. 13 ✓

d. 16

Substituting $x = 2$ into $y = 3x + 7$ gives $y = 3(2) + 7 = 13$.

Graphical & Geometric Interpretation

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

7 In $\triangle ABC$ studying review: identifying congruent triangles, $m\angle A = 30^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 91° ✓

b. 101°

c. 83°

d. 150°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (30^\circ + 59^\circ) = 91^\circ$.

8 A geometric figure scaled by factor $k = 4$ has an original side length of 4 cm. What is the corresponding image length?

a. 12 cm

b. 16 cm ✓

c. 20 cm

d. 64 cm

Multiplying the original length by the scale factor gives $4 \times 4 = 16$ cm.

9 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 7)$. What is its y-coordinate when $x = 3$?

a. 13

b. 18

c. 16 ✓

d. 19

Substituting $x = 3$ into $y = 3x + 7$ gives $y = 3(3) + 7 = 16$.

Exam-Style Problem Solving & Modeling

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

10 In $\triangle ABC$ studying review: identifying congruent triangles, $m\angle A = 30^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 84° ✓

b. 94°

c. 76°

d. 150°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (30^\circ + 66^\circ) = 84^\circ$.

11 A geometric figure scaled by factor $k = 5$ has an original side length of 4 cm. What is the corresponding image length?

a. 15 cm

b. 20 cm ✓

c. 24 cm

d. 100 cm

Multiplying the original length by the scale factor gives $4 \times 5 = 20$ cm.

12 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 7)$. What is its y-coordinate when $x = 4$?

a. 16

b. 21

c. 19 ✓

d. 22

Substituting $x = 4$ into $y = 3x + 7$ gives $y = 3(4) + 7 = 19$.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647